

Air Quality Data Cleaning Report by Vamshi Aitharaju

Data Description

My dataset is about Air quality and Health Impacts. This data was provided to public by NYC Department of Health and Mental Hygiene. This is URL for the original Dataset here ([Link](#)). This data set was collected to show how environments affect health, by publishing data and information that explain the connection pollution and causes. This dataset includes neighborhood averages of different air pollutants, estimates of health impacts of air pollution. So, my data set provides information about pollutants (NO2, Ozone, PM2.5) measurement definitions (Units), geographic coverage (different division) based on different seasons and annual average of pollution created. My data set also consist of Meta data like data about data, for example measure and measure info. Initial data was collected between 2005 to 2023 along with separate seasons and also annual average. The final updated data was on 4/2/2024 adding December 2021, January 2022 and June 2022 data. So, while I was going through the Link provided initially, I got information the how Air Pollutant Value was calculated, All the Air samples collected at specific NYCCAS monitoring sites. Annual mean is the average for 12 months, summer mean is June-August, and winter mean is December-February, marked by the year it ends.

Data Dictionary:

Data Dictionary		
Column Name	Description	Type
Index	Record identifier	Identifier
Name Of the Pollutant	Pollutant name	Categorical
Measure	Statistical measure type (Mean, Annual Average)	Categorical
Measure Info	Unit of measurement (PPB,mcg/m3)	Categorical
Geo Type Name	Geographic level (CD, UHF42, Borough, Citywide)	Categorical
Geo Place Name	Specific geographic location name	Categorical
Time Period	Label describing time (Winter 2014-15, Annual Average 2017)	Categorical/Temporal
Start Date	Exact start date for the period	Temporal (date/time)
Air Pollutant Value	Observed concentration of pollutant	Quantitative

Approach and method to Used to Clean the data

First thing I did after opening my data set is I tried to read and understand the data at a quick glance. I read all the variables and what kind of categories, how many categories and qualitative columns were Present along with units and dates, so that it will give an idea to form business question and what and where to focus. So, when I saw it was a huge data set with 18k observations. Surprisingly, I have observed that there were no null values, no empty cells except one empty column with data label Message. That's where I have realized I have to segregate the data before I clean it because there is not much to clean at initial stage and also there were like multiple components such as distance travelled by vehicles, deaths, PM2.5 which was confusing. That's where I realized that I have to focus on one particular thing. So, what I did was I have created a pivot chart and studied all components and the count, there I found out that I can categories all those names into 3 different categories pollutants, vehicles, deaths and hospitalizations. So, I used power query and created 3 different sheets based on categories. So I thought I was find and can proceed with data cleaning But again there was a doubt raised as there are multiple pollutants, deaths and vehicles miles travelled with different measurement units, so it is like comparing apples vs oranges, so again I have looked at Pivot table and understood that I have a scope to focus more on PM2.5, Ozone and NO2 as they are main reason for all the other 2

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categories(Deaths, emissions caused by vehicles) and added advantage is that they have very good no of observation which leads a good data accuracy, so I have created one single sheet with PM2.5, Ozone and NO2 and then again face same issue because the years are so confusing with different types like it has 2005, winter 2008-09, summer 09 and then annual average 09, so I again I spend some time and I removed the outliers like 2005, 2007 as they going to not add any value to the data and it is not in sequence, so I took the most common started years 2009-2023, Seasons and Annual average. Then after coming to geo type, I had different component, so i researches in google for geo type in NYC, learned about borough, CD, UHF, Citywide, they cover same area but different divisions of area, like CD was divided into 56 districts where UHF34 into 34, UHF42 into 42 and city wide as a whole. I selected CD as it is suitable for my analysis, as because even any Geo-type covers same city but the divisions matter as I can use it to point out exactly where there is high pollution and its cause pollutant. Then after applying CD filter, Data set was clear and easier to understand at a glance, then I have started cleaning the data set in power query. I have Removed unnecessary columns such as Indicator ID, Geo Join ID and message column as they are not that useful it is all meta data. I removed duplicates, replaced unique Id with my own Index Id for convenience, removed duplicates, Renamed Data value as Air Pollutant Value based on my selected components in the data set, Changed the date format, rounded off the Air Pollutant Value, Filtered Years, Given Abbreviation names such as CD as Community District, PPB as Parts Per Billion for better understanding. Then after my data set was ready. Then after as per Professor Todd's and Professor Kristin's insight, I have created three different sheets based on summer season, Winter Season and Annual average. So, it did not took much time to clean the data until and unless I understood my data variables like Units, Pollutants, Geotype.

discrepancies or anomalies

1. At very first glance I have observed an empty column completely (Message Column)
2. Raw data was collected and mixed up with all components such as death due to PM2.5, Annual Miles travel by Vehicles and pollutants with different measure and measure info along with different time period.
3. Time Period Inconsistencies observed in the data set, few samples were collected as in 2005, 2005-2007. Which looks like outliers and cannot be used for analysis.
4. As the Data was collected annually, we only have average of winter, summer and annual but not monthly value, Air pollutant Value is average of particular season and year's average.
5. Start Date for Winter and summer is being same for whole data set as it was calculated annually, but it was different for annual average
6. I have observed Unit differences because different pollutant is measures in different unit like ppb, mdg/m3 and other units, so I was bit confusing and I have segregated in such a way that, I understood and I can let other understood it.
7. Geo Type was most difficult part for me, because there 5 different types where all 5 covers same New York City but in different type of divisions. So, I took a bit of research to figure out what exactly I want and what exactly is useful for analysis.

Here by I conclude that, the Data cleaning has been done by Vamshi Aitharaju and it is ready for review.